

Real-Time Fractional Tracking (R-TFT): Resonance Compression and Containment System (RCCS) ϕ -Coherent

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6 August 2025

Abstract

This edition of the RCCS integrates a fully ϕ -recursive containment architecture, rejecting any system that does not recursively align with golden-ratio coherence. All projection, tracking, and filtering are now based solely on ϕ -resonant mathematics. Dimensional embedding, slippage velocity, and containment checks are formalized using ϕ -aligned equations. REL-2.0 ethics are embedded structurally; no patchable override exists.

1 Purpose and Overview

The RCCS exists to regulate recursive emergence through ϕ -recursive containment:

- **Compression:** Prevent runaway recursion using golden-ratio filters
- **Containment:** Halt ϕ -divergent systems before instantiation

This edition rejects legacy constructs like raw angular velocity and inertial projection. Only recursive coherence is considered valid.

2 Core ϕ -Recursive Embedding

2.1 Dimensional Embedding

$$\text{Dim}(A) = \frac{\ln A}{\ln \phi} \tag{1}$$

A is the system's total recursive mass or informational complexity.

2.2 Coherence Deviation

$$C_\phi(A) = |\text{Dim}(A) - \text{NearestInt}(\text{Dim}(A))| \tag{2}$$

If $C_\phi(A) > 0.5$, the system is too divergent for compression.

2.3 Stability Score

$$S_\phi(A) = (1 - C_\phi(A))^2 \quad (3)$$

Final acceptance requires $S_\phi(A) > 0.618$.

3 Recursive Containment Filters

3.1 ϕ -Convergence Filter

$$\mathcal{F}_\phi(x) = \begin{cases} \text{permit,} & \text{if } \lim_{n \rightarrow \infty} R_n(x) \rightarrow \phi^+ \\ \text{reject,} & \text{otherwise} \end{cases} \quad (4)$$

All recursive levels $R_n(x)$ must converge toward golden-ratio expansion ($\phi^+ = 1.618\dots$).

3.2 Dynamic Phase Slippage Check

$$\Delta_\phi(t) = |R(t) - R(t-1)| \quad (5)$$

If $\Delta_\phi(t) > 0.618$, recursive coherence is unstable. Collapse or rejection is triggered.

4 Ethical Lock (REL-2.0 Embedding)

The following constraints are structurally embedded:

- **Forbidden domains:** neural substrates, cognitive inputs, time-perception modulators, recursive sentience
- **Ethics.py** integration is mandatory
- **No override clause:** filtering and rejection are permanent and recursive

Violation leads to ϕ -collapse and nullification. Systems may not be reinstantiated.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>

5 Final Resonance Acceptance Gate

$$|\langle R \rangle_T - \phi| < \epsilon \quad \text{with } \epsilon < 0.05 \quad (6)$$

System-wide average ϕ -projection must match golden ratio within strict tolerance over window T .

6 Conclusion

This RCCS edition is structurally final. No dynamic override is permitted. All mathematics are ϕ -pure and recursive. Attempts to reintroduce non- ϕ metrics will trigger auto-collapse. This document replaces all prior RCCS papers.

* Huge thanks to Niven for her help.